PROJECT STAGE – I Report

On

BLOCKCHAIN BASED MICRO CREDIT SYSTEM FOR FREELANCERS

Submitted in partial fulfilment of the requirements for completion of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND BUSINESS SYSTEMS

by

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(Affiliated to JNTUH, Hyderabad; Eight UG Programs Accredited by NBA; Accredited by NAAC with 'A++' Grade)

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CERTIFICATE

This is to certify that the Project Stage - I entitled **"BLOCKCHAIN BASED MICRO CREDIT SYSTEM FOR FREELANCERS",** is being submitted by **K. Tarun bearing Roll No: 22261A3228 and B. Aishwarya bearing Roll No: 22261A3202** in partial fulfilment of completion of **Bachelor of Technology VII Semester** in **Computer Science and Business Systems** to **Mahatma Gandhi Institute of Technology** is a record of bona-fide work carried out under the supervision of **Dr. M. Rudra Kumar**. The results embodied in this project have not been submitted to any other University or Institute for the award of degree or diploma.

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DECLARATION

This is to certify that the work reported in Project Stage - I titled "**BLOCKCHAIN BASED MICRO CREDIT SYSTEM FOR FREELANCERS"** is a record of work done by us in the Department of Computer Science and Business Systems, Mahatma Gandhi Institute of Technology, Hyderabad, under the guidance of **Dr. M. Rudra Kumar, Professor**, Department of IT, MGIT. No part of the work is copied from books/journals/internet and wherever the portion is taken, the same has been duly referred in the text. The report is based on the work done entirely by us and not copied from any other source.

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ABSTRACT

Freelancers and gig-economy workers face significant challenges when accessing formal credit due to inconsistent income patterns, limited documentation, and the absence of traditional credit histories. Conventional financial systems are not designed to evaluate digital contributors who work across platforms and countries. This creates an urgent need for alternative, technology-driven credit solutions tailored to their financial realities.

This project introduces a blockchain-based micro-credit system that leverages decentralized identity, alternative data sources, and AI-powered credit scoring to create a transparent and trustworthy lending framework. Gig-platform performance metrics, payment histories, and digital behavioral indicators are aggregated to construct a reliable borrower profile. Smart contracts automate lending workflows by handling loan approval, disbursement, and repayment without intermediaries, ensuring security and tamper-proof execution.

The system maintains a portable on-chain credit identity that grows stronger as freelancers repay loans, enabling them to access financial support across borders and platforms. By reducing processing overhead, improving transparency, and leveraging decentralized trust mechanisms, the solution creates a scalable micro-lending ecosystem. Ultimately, this framework enhances financial inclusion by enabling millions of independent workers to access fair, secure, and affordable credit opportunities.

1. INTRODUCTION

The rise of the gig economy has created new income opportunities for freelancers, yet their access to formal credit remains extremely limited. Traditional lending frameworks rely on stable salaries and long-term financial records, which freelancers often do not possess. This mismatch leaves a large segment of the workforce financially underserved.

Advances in blockchain and machine learning now offer the ability to redesign credit assessment for non-traditional workers. Blockchain provides a secure and transparent infrastructure for identity verification and loan tracking, eliminating the need for intermediaries. Machine learning enables lenders to evaluate creditworthiness using alternative and behavioral data sources.

This project develops a blockchain-based micro-credit system tailored for freelancers with irregular income streams. It aggregates digital payment history, gig-platform performance, and behavioral indicators to generate a reliable credit profile. Smart contracts automate loan approval, disbursement, and repayment, ensuring trustless execution.

The goal of this system is to empower freelancers with fair, secure, and accessible micro-credit opportunities. By offering a portable on-chain credit identity, the solution supports cross-platform verification and long-term financial inclusion. Ultimately, it creates a scalable and transparent lending ecosystem for the digital workforce.

1.1 Background

The rise of the gig economy has created a large and fast-growing workforce that operates outside traditional employment structures, resulting in limited access to formal credit and financial services. Freelancers often lack stable salaries, employer verification, and conventional credit histories, making them invisible to traditional lending systems. Advances in blockchain and machine learning provide an opportunity to build transparent, data-driven, and decentralized lending models that can evaluate borrowers using alternative digital indicators. This motivates the development of a blockchain-based micro-credit system that securely assesses freelancer creditworthiness and enables fair, scalable, and inclusive lending.

1.2 Motivation

Freelancers are increasingly contributing to the digital economy, yet they remain largely excluded from formal credit due to irregular incomes and lack of traditional financial records. This creates a need for alternative, technology-driven lending models that recognize their digital earning patterns. With blockchain and AI offering secure and transparent evaluation methods, there is strong motivation to build a system that enables fair, accessible, and trust-based micro-credit for freelancers.

1.3 Problem Statement

Freelancers and gig-economy workers struggle to access formal credit because traditional lending systems depend on stable income, employer verification, and conventional credit histories—none of which align with the realities of platform-based work. As a result, this growing workforce segment is systematically excluded from financial services despite having consistent earnings through digital platforms.

There is a critical need for a secure, transparent, and data-driven lending mechanism that can evaluate freelancer creditworthiness using alternative indicators such as digital payments, gig-platform performance, and behavioral patterns. Existing systems lack decentralized trust, automation, and reliable identity verification, creating inefficiencies and risks in micro-lending. This project aims to address these gaps by building a blockchain-based micro-credit system that enables fair, automated, and inclusive credit access for freelancers.

1.4 Existing System

The existing financial ecosystem for freelancers is largely fragmented and exclusionary, offering little to no support for individuals lacking stable income documentation or credit histories. Traditional banks depend on CIBIL-based scoring, employer verification, and collateral requirements, none of which align with the flexible income structures of gig workers. As a result, freelancers often resort to informal, high-interest lenders with little transparency or protection. Existing digital lending platforms also lack decentralized trust mechanisms, rely on shallow data for risk assessment, and provide limited automation. This leads to slow approvals, high rejection rates, and inaccurate credit decisions.

1.5 Proposed System

The proposed blockchain-based micro-credit system addresses these limitations by leveraging decentralized identity, alternative data aggregation, and AI-powered credit scoring. The system collects gig-platform earnings, digital payment patterns, and behavioral indicators to build a reliable financial profile for freelancers. Smart contracts automate loan approval, disbursement, repayment, and contract enforcement, ensuring transparency and eliminating intermediaries. A portable on-chain credit identity enables freelancers to carry verified financial records across platforms and geographies. Overall, the system provides a secure, scalable, and fair micro-lending framework that improves financial inclusion for the global freelance workforce.

2. LITERATURE SURVEY

[1] Wang & De Filippi (2020), "Self-Sovereign Identity in a Globalized World"

This paper presents a foundational exploration of Self-Sovereign Identity (SSI) as a decentralized framework that empowers individuals to control their digital identities without reliance on centralized authorities. The authors highlight how blockchain ensures verifiable credentials, cross-border trust, and enhanced privacy preservation. Their analysis illustrates SSI's ability to solve identity fragmentation and reduce administrative overhead for vulnerable populations. The study emphasizes that global interoperability remains a significant hurdle, especially due to differing regulatory landscapes and lack of standardized credential formats.

[2] Grech (2021), "Blockchain, Self-Sovereign Identity & Digital Credentials"

Grech investigates how blockchain can streamline digital credential management using verifiable, tamper-proof identity proofs. The work focuses on educational and governmental use cases, demonstrating how SSI reduces fraud and enhances data portability across institutions. The study underscores the scalability advantages of decentralized identity compared to traditional certificate repositories. However, the paper acknowledges persistent barriers related to technology adoption, data governance policies, and institutional readiness for decentralized credential ecosystems.

[3] Vasselin (2024), "Blockchain for Migrants: Promoting SSI & Financial Inclusion"

This paper explores the use of blockchain-based SSI systems for migrants who often lack formal identity documents and financial access. Vasselin proposes a prototype demonstrating how decentralized credentials improve access to essential services such as banking, healthcare, and employment verification. The findings indicate strong potential for blockchain to reduce discrimination and streamline cross-border identity portability. Nonetheless, the study is limited by small-scale testing and highlights the need for real-world deployment data to validate long-term feasibility.

[4] Isreal (2025), "Future of Credit Scoring: Blockchain, ML & Alternative Data"

Isreal introduces an integrated framework combining blockchain with machine learning to redesign the traditional credit scoring lifecycle. The paper emphasizes the use of alternative data—such as digital transactions, behavioral patterns, and platform activity—to generate more inclusive borrower assessments. By storing audit trails and consent-driven data access on-chain, the model boosts transparency and fairness in lending. Although promising, the study notes the absence of large-scale benchmarking on real financial datasets, which limits immediate practical deployment.

[5] Onerhime & Ehiedu (2025), "Blockchain-Based Credit Scoring in Developing Economies"

This work proposes a hybrid blockchain credit scoring model targeted at underserved populations within developing countries. The authors demonstrate how decentralized ledgers reduce fraud, increase lender confidence, and enable transparent verification of borrower histories. The system leverages a mix of on-chain identity proofs and off-chain behavioral metrics to generate credit insights. The paper acknowledges, however, that limited empirical validation in live markets and infrastructural challenges may hinder widespread adoption.

Table 2.1: Literature Survey Table

Author	Title	Year	Name of the Journal	Methodology / Concept Used / Summary	Merits	Demerits
Wang & De Filippi	Self-Sovereign Identity in a Globalized World	2020	Frontiers in Blockchain	Decentralized identity for global user verification	SSI enhances privacy, control, inclusion	Limited empirical validation; scalability
Grech	Blockchain, Self-Sovereign Identity & Digital Credentials	2021	Frontiers in Blockchain	Blockchain-based tamper-proof digital credentials	Tamper-proof verifiable identity	Adoption barriers; regulatory complexity
Vasselin	Blockchain for Migrants & Financial Inclusion	2024	Preprints	SSI-driven financial access for migrants	Improves migrant financial access	Prototype; lacks deployment evidence
Isreal	Future of Credit Scoring: Blockchain & ML	2025	ResearchGate Working Paper	No real banking dataset	Blockchain+ML improves inclusion	No real banking dataset
Onerhime & Ehiedu	Blockchain Credit Scoring	2025	IIARD Journal of Banking & Finance	Hybrid blockchain model for trustworthy credit	Trust ↑, fraud ↓ in lending	Limited live testing

3. SYSTEM SPECIFICATIONS

3.1 Software Requirements

The technologies outline the core development tools, frameworks, and platforms utilized to build and integrate all components of the Blockchain-Based Micro-Credit System for Freelancers. This includes web technologies for creating an intuitive user interface, backend frameworks for secure API handling, blockchain tools for smart contract management, and machine learning libraries for alternative credit scoring. Together, these technologies form a cohesive and scalable ecosystem that supports decentralized identity, transparent loan processing, data-driven scoring, and seamless user interaction across devices.

- **Frontend Stack:**
    - ReactJS – Used to build a responsive, component-driven user interface for borrower onboarding, loan applications, and repayment dashboards.
    - Tailwind CSS / Bootstrap for styling – Enables fast, consistent UI styling with mobile-first design principles, improving usability across devices.
- **Backend Stack:**
    - Node.js (v18 or later) - Provides a scalable backend runtime for API execution, blockchain interactions, and loan-processing logic.
    - Express.js (v4 or later) - Facilitates modular REST API development, including routing, middleware integration, and service communication.
    - JWT for authentication - Used for secure user login and session validation, ensuring protected access to loan services and identity modules.
- **Blockchain & Smart Contract Stack:**
    - Solidity (for Smart Contracts) - Implements loan agreements, repayment logic, and identity anchoring on an EVM-compatible blockchain.
    - Hardhat / Truffle Framework - Supports smart contract compilation, migration, testing, and debugging in local and testnet environments.
    - Ethers.js / Web3.js - Enables secure interaction between the backend/frontend and the blockchain for contract deployment and transaction handling.

    - Ganache / Local EVM Node - Used for local blockchain simulation during development and testing.
- **Machine Learning & Analytics Stack:**
    - Python 3.8+ - Used to implement ML-based credit scoring using alternative datasets such as gig-platform performance and payment behavior.
    - Flask API / FastAPI - Serves the ML inference pipeline, enabling scoring requests from the backend system.
    - Scikit-learn (Random Forest, Gradient Boosting) - Used for credit risk classification and score generation based on engineered behavioral features.
    - TensorFlow / PyTorch (for LSTM Forecasting) - Supports optional income forecasting models to analyze freelancer earning patterns.
- **Database**: MongoDB (local or MongoDB Atlas) - Used to store user profiles, loan applications, repayment logs, and alternative data securely and efficiently.

3.2 Hardware Requirements

- Operating System: Windows 10/11 (64-bit), Ubuntu 20.04+
- IDE/Text Editor: Visual Studio Code / PyCharm - Used as primary IDEs for coding, debugging, and collaborative development across frontend, backend, ML, and blockchain modules.
- API Integrations:
    - UPI/Payment Gateway APIs (Razorpay/Paytm/Stripe) - Integrated to facilitate real-time loan disbursement and repayment collection.
    - Gig-Platform APIs (Optional) - Support importing freelancer earnings, ratings, and job history for credit scoring.
- Package Managers:
    - npm (for JavaScript dependencies) - For installing Node.js dependencies and smart contract toolchains.
    - pip (for Python packages) - For managing Python packages used in ML pipelines.

- Internet Connection: Required for blockchain node syncing, API calls, gig-platform data ingestion, and cloud deployment.

Development Machine

- CPU: Intel i5 / AMD Ryzen 5 (Quad-core or higher)
- RAM: Minimum 8 GB (16 GB recommended for blockchain & ML)
- Storage: At least 20–30 GB free
- Display: 1366×768 resolution or higher
- GPU (Optional): NVIDIA CUDA-enabled GPU for ML acceleration

Target Deployment Setup (Optional - Cloud/Edge Devices)

- Cloud Hosting: AWS EC2 / Render / DigitalOcean / GCP (backend + blockchain node)
- Database Hosting: MongoDB Atlas / PostgreSQL Cloud
- Blockchain Node: Infura, Alchemy, or self-hosted Quorum/Polygon node
- Client Compatibility: Chrome, Edge, Firefox (latest versions)

4. DESIGN AND METHODOLOGY

4.1 System Design

The system is organized into modular components, each responsible for a distinct function within the micro-credit lifecycle. These modules handle identity verification, data aggregation, credit scoring, loan contract execution, payments, and user management. All modules communicate through secure REST APIs and blockchain transactions, ensuring loose coupling, scalability, and fault tolerance across the platform.

4.1.1 Module Description

1. User Portal Module:

- Serves as the main interface for freelancers to onboard and manage loan applications.
- Provides access to KYC submission, credit score viewing, loan status, and repayment updates.
- Eliminates the need for complex login systems when using blockchain wallet authentication.
- Communicates with backend services through secure API calls.

2. Identity & KYC Verification Module:

- Handles user identity creation, verification, and mapping to decentralized identifiers (DIDs).
- Integrates with external KYC providers for document verification and risk checks.
- Stores verified identity in an off-chain database while writing identity anchors to the blockchain.
- Prevents fake or duplicate users through strict validation.

3. Data Aggregation Module:

- Collects alternative financial data such as gig-platform earnings, payout cycles, and ratings.
- Aggregates digital payment history fetched from APIs or uploaded records.

- Normalizes and stores data in a dedicated financial datastore.
- Ensures input validation to block fraudulent or incomplete data sources.

4. Credit Scoring & Risk Engine:

- Generates credit scores using hybrid ML models trained on alt-data, consistency, and historical patterns.
- Uses features from gig-platform performance, digital transaction behavior, and reliability indicators.
- Produces a risk tier, recommended loan amount, interest rate, and approval probability.
- Stores scoring components and timestamps in the database for auditability.

5. Smart Contract Loan Module:

- Deploys blockchain-based loan contracts defining principal, tenure, interest, penalties, and repayment rules.
- Automates loan activation, disbursement triggers, repayment updates, and default marking.
- Ensures tamper-proof, transparent loan lifecycle without intermediaries.
- Logs on-chain events corresponding to repayments and contract state transitions.

6. Payment & Settlement Module:

- Executes loan disbursement and repayment via UPI or bank rails.
- Integrates with a payment gateway for real-time fund movement and confirmation.
- Reconciles off-chain payments with on-chain loan contract state.
- Handles partial payments, transaction failures, and settlement retries.

6. Reputation & Incentive Module:

- Maintains a reputation score based on repayment discipline and platform engagement.
- Rewards reliable users with lower interest rates and higher future credit limits.

- Updates reputation ledger periodically and stores it in the database.
- Helps lenders identify trustworthy borrowers in peer-to-peer lending environments.

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- Maintains a reputation score based on repayment discipline and platform engagement.
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- Updates reputation ledger periodically and stores it in the database.
- Helps lenders identify trustworthy borrowers in peer-to-peer lending environments.

6. Admin Dashboard & Monitoring Module:

- Allows administrators to review loan applications, risk scores, and system analytics.
- Provides audit tools, compliance checks, and contract lifecycle monitoring.
- Displays disbursement, default, and repayment statistics in real time.
- Enables override controls such as manual approval or intervention.

4.1.2 Architecture of Proposed System

Architecturally the system is layered: a responsive Frontend (web/mobile) interacts with an API Gateway that routes requests to microservices—Onboarding & DID issuance, Data Ingestion (gig/payments), Alternative-Data Profiling, ML Scoring & Forecasting, Smart-Contract Orchestrator, Payment/Settlement, Reputation Engine, and Admin/Risk UI. Persistent stores include a transactional DB for users/loans, a feature store/Redis for real-time inference, and a blockchain ledger for identity anchors and on-chain loan contracts. Auxiliary services provide KYC/AML, external API connectors (gig platforms, payment rails), observability, and secrets management to ensure secure, auditable, and scalable operation.

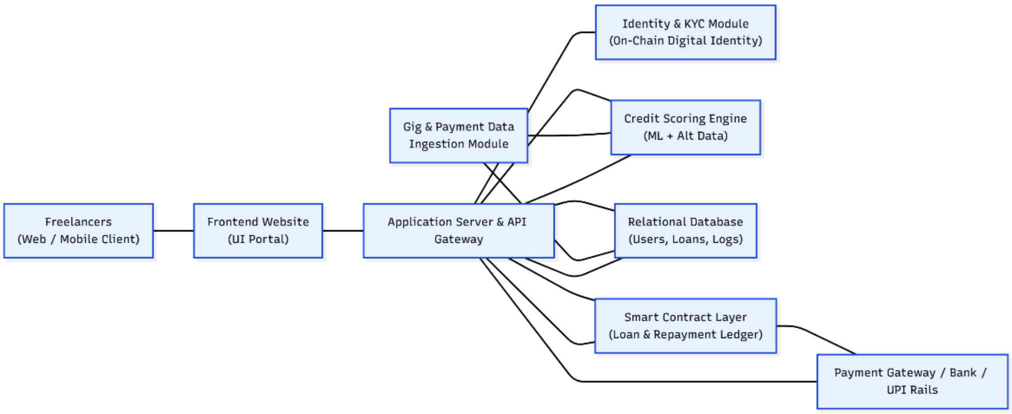

Fig 4.1.2 Architecture of Proposed System

4.2 Methodologies

- **Data Collection**: The system collects detailed information about freelancers, including their income patterns, project history, client ratings, and transaction records. This data acts as the foundation for evaluating their financial stability and repayment capacity.
- **Credit Scoring**: Machine Learning models are employed to analyze the collected data and determine each applicant's creditworthiness. These models predict the probability of repayment or default, enabling fair and data-driven loan decisions.
- **Smart Contracts**: Blockchain-based smart contracts are deployed to automate the lending process. They handle the entire loan lifecycle — from application and approval to disbursal and repayment — ensuring transparency and eliminating the need for intermediaries.
- **Reputation System**: A dynamic reputation score is maintained for every freelancer. It is automatically updated after each successful repayment, creating a trustworthy credit history that improves access to future loans.
- **Security & Transparency**: Blockchain technology ensures all transactions are recorded immutably, providing a tamper-proof ledger that guarantees data integrity, transparency, and verifiable user identity.

4.3 UML Diagrams

The detailed design outlines how internal services, smart contract components, and external data sources interact to deliver secure micro-credit functionality. The backend exposes essential endpoints such as /api/kyc/verify, /api/loans/apply, /api/score/:userId, and /api/repay/:loanId, each responsible for orchestrating blockchain transactions and alternative-data processing.

These API routes communicate with Python-based credit scoring services, data aggregation modules, and the smart contract handler to generate reliable loan decisions. User profiles, KYC records, financial activity, and repayment histories are stored in the database using schemas optimized for identity, loan status, alternative-data features, and on-chain references.

All validated results are then passed back to the frontend dashboard, enabling freelancers to access loan eligibility, contract details, and repayment insights in real time.

4.3.1 Use Case Diagram:

The Use Case Diagram outlines the primary interactions between system actors—such as freelancers, admins, the payment gateway, and the blockchain network—and the core functionalities they access. It highlights essential user operations like onboarding, KYC verification, loan application, and repayment. The diagram provides a high-level functional overview by mapping actor-driven actions to system capabilities. This ensures clarity on what each actor can initiate or influence within the platform.

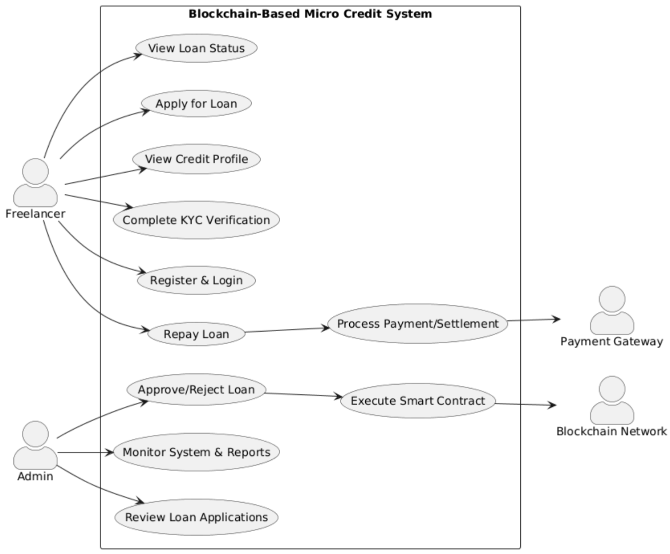

Fig 4.3.1 Use Case Diagram

4.3.2 Sequence Diagram:

The Sequence Diagram illustrates the chronological flow of operations between the user, frontend portal, backend services, smart contract layer, and payment gateway. It depicts how requests such as loan application, credit scoring, contract creation, and repayment are exchanged step-by-step. This view helps clarify the timing, sequencing, and dependencies between modules in real-time execution. It ensures the end-to-end system workflow is transparent and logically aligned.

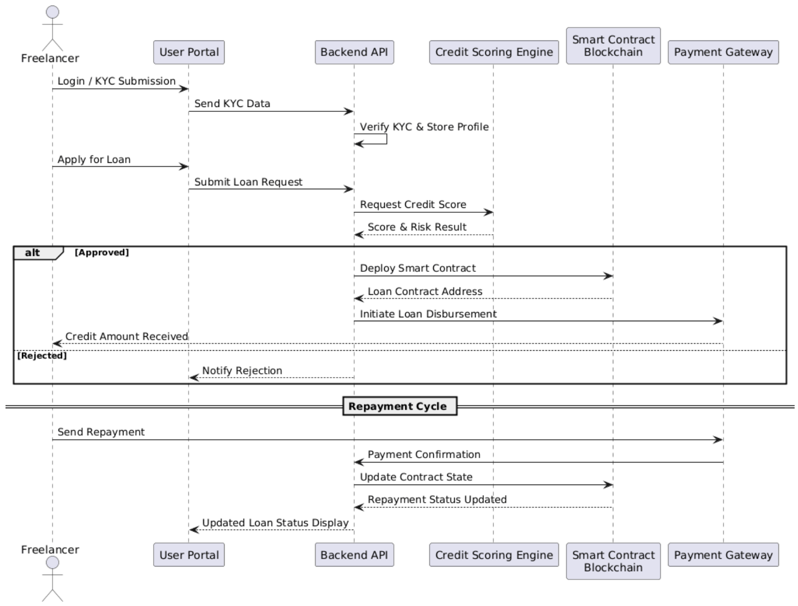

Fig 4.3.2 Sequence Diagram

4.3.3 Activity Diagram:

The Activity Diagram models the procedural flow of the loan lifecycle—from user login and KYC verification to loan approval, contract execution, repayment, and closure. It highlights conditional decisions such as approval checks, repayment loops, and default handling. The diagram captures the dynamic behavior of the system through interconnected actions and decision paths. This representation helps validate functional logic and process coherence.

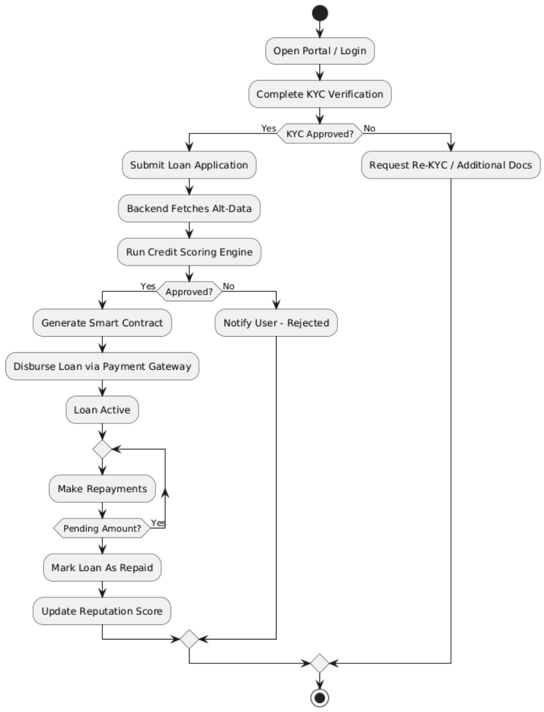

Fig 4.3.3 Activity Diagram

4.3.4 Class Diagram:

The Class Diagram defines the system's structural blueprint by detailing key classes like User, KYC, LoanApplication, CreditScore, Payment, and SmartContract. It outlines class attributes, core methods, and the relationships between entities such as associations and compositions. This structure supports modularity, data integrity, and extensibility across backend components. It serves as the foundation for object-oriented system design and implementation.

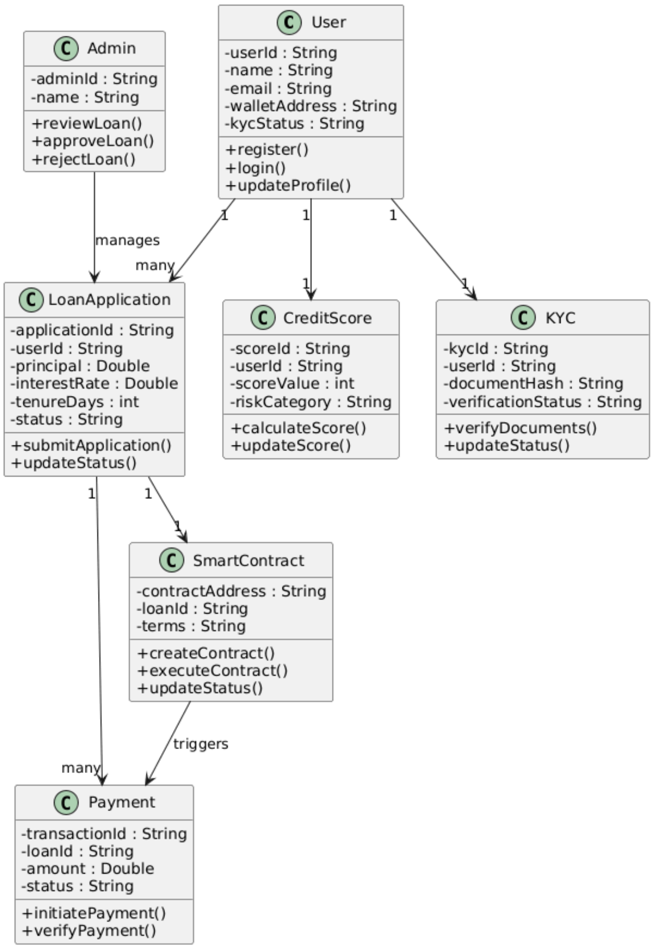

Fig 4.3.4 Class Diagram

4.3.5 Component Diagram:

The Component Diagram illustrates the system's high-level architecture by dividing it into frontend, backend API, identity services, ML modules, blockchain handler, and payment integration components. It showcases how these modules communicate through REST APIs and external connectors. The diagram emphasizes modularity, separation of concerns, and scalability across the platform. It provides a holistic view of system-level interactions and deployment responsibilities.

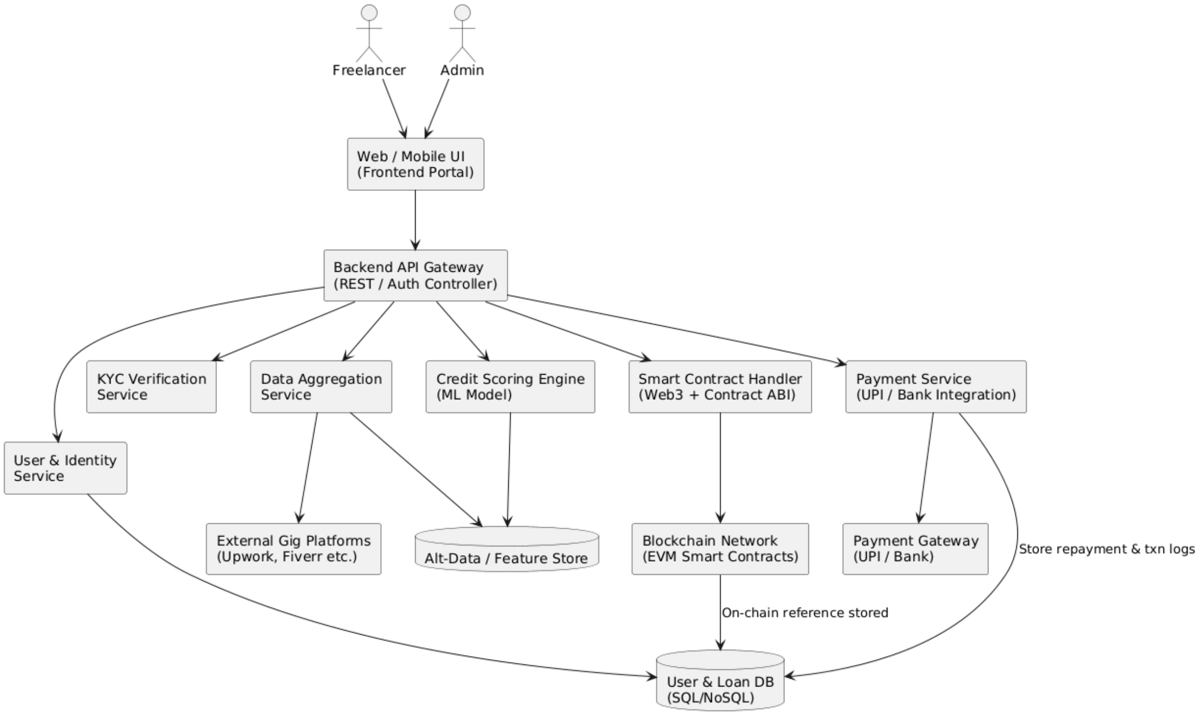

Fig 4.3.5 Component Diagram

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